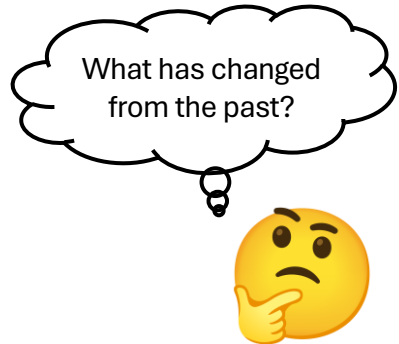
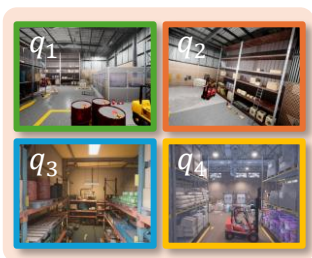
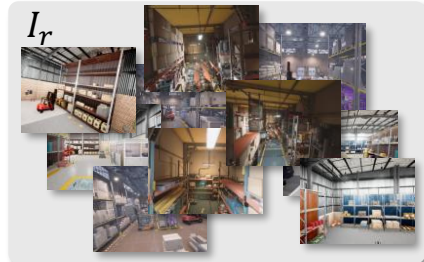




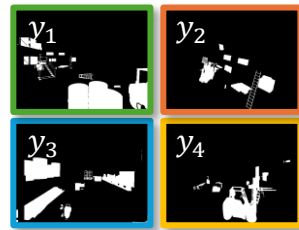
Query
(present scene)



Reference image Database
(past environment)

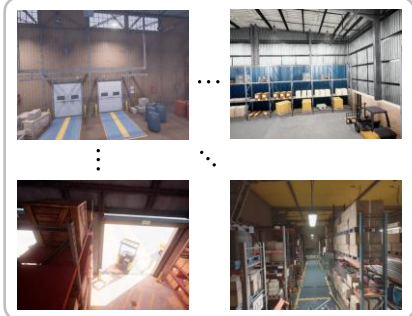


Changes in query image vs.
the reference environment



1. Reference Subset Construction

Reference Database $I_r = \{r\}$



Query Image q



Subset
Constructor

Reference Subset $\mathcal{R}$



2. Reference Mosaicing

Feature
Extractor

$\{f_r^{(k)}\}$

Environment
Mosaic
Composer

Feature
Extractor

f_q

Semantic
Aggregator

f_r^*

$\tilde{f}_r$



Change
Detector

3. Semantic Aggregation

